

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : Cyber & Information Security

AIM : Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

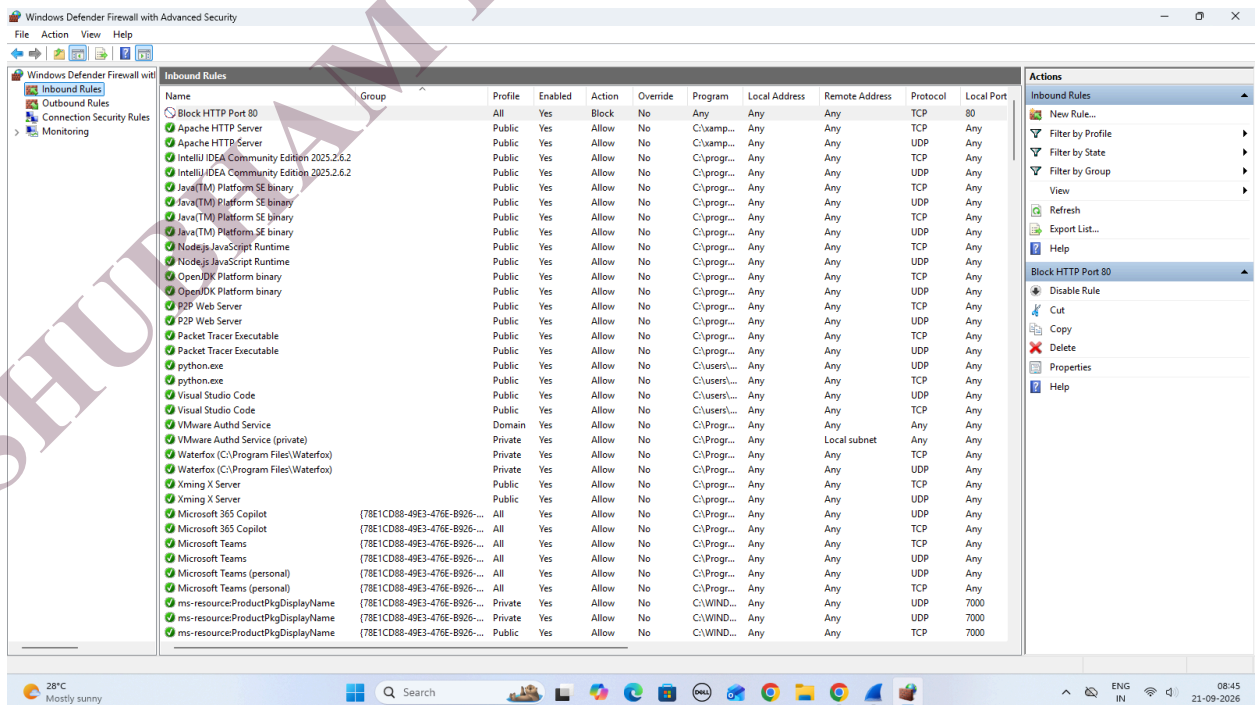
Steps:

1. Open Windows Firewall

1. Press Windows + R.
2. Type: wf.msc
3. Press Enter.
4. Windows Defender Firewall with Advanced Security will open.

2. Create an Inbound Firewall Rule

1. Click **Inbound Rules** on the left.
2. Click **New Rule...** on the right.
3. Select **Port** → **Next**.
4. Select **TCP**.
5. Enter a port, for example: 80
6. Click **Next**.
7. Select **Block the connection**.
8. Click **Next**.
9. Select **Domain, Private, Public**.
10. Give the rule a name: Block HTTP Port 80



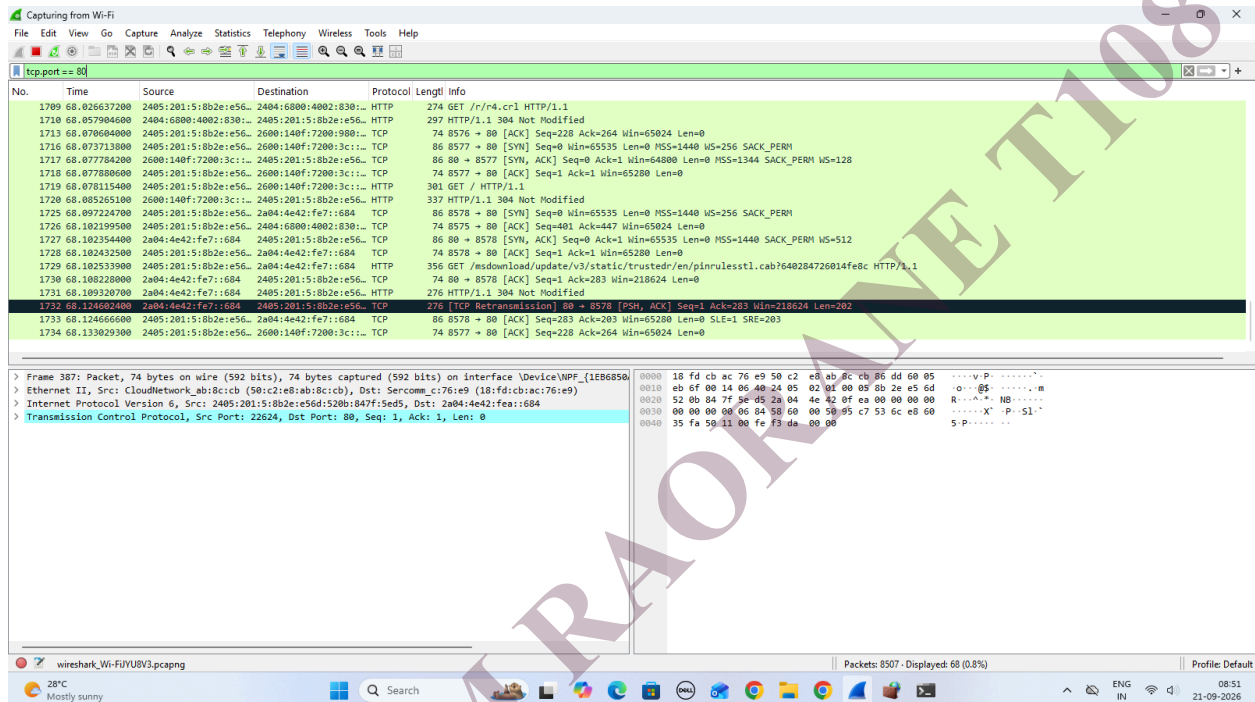
11.

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3. Start Wireshark

1. Open **Wireshark**.
2. Select your active network interface:
 - Wi-Fi, or
 - Ethernet
3. Click **Start Capturing Packets**.



4. Filter Packets in Wireshark

To view HTTP traffic: **tcp.port == 80**

To view TCP traffic: **tcp**

To view traffic from a particular IP: **ip.addr == 192.168.1.10**

To view blocked-port traffic: **tcp.port == 80**

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Capturing from Wi-Fi

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
97.222	192.168.29.120	TCP	1514 443 → 59519 [ACK] Seq=18839 Ack=3176 Win=4193824 Len=1460 [TCP PDU reassembled in 379]			
97.222	192.168.29.120	TLSv1.3	1261 Application Data			
29.120	204.79.197.222	TCP	54 59519 → 443 [ACK] Seq=3207 Ack=20299 Win=65280 Len=0			
29.120	204.79.197.222	TCP	54 59519 → 443 [ACK] Seq=3207 Ack=21586 Win=64256 Len=0			
97.222	192.168.29.120	TCP	1514 443 → 59519 [ACK] Seq=21586 Ack=3176 Win=4193824 Len=1460 [TCP PDU reassembled in 383]			
97.222	192.168.29.120	TLSv1.3	318 Application Data			
97.222	192.168.29.120	TCP	54 443 → 59519 [ACK] Seq=23261 Ack=3207 Win=4193824 Len=0			
29.120	204.79.197.222	TCP	54 59519 → 443 [ACK] Seq=3207 Ack=23230 Win=65280 Len=0			
15:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0			
15:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 22626 → 80 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0			
2:fea1:684...	2405:2015:8b2e:e56...	TCP	74 80 → 22626 [ACK] Seq=1 Ack=2 Win=427 Len=0			
2:fea1:684...	2405:2015:8b2e:e56...	TCP	74 80 → 22626 [FIN, ACK] Seq=1 Ack=2 Win=427 Len=0			
15:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 22626 → 80 [ACK] Seq=2 Ack=2 Win=255 Len=0			
29.120	204.79.197.222	TCP	54 59519 → 443 [ACK] Seq=3207 Ack=23261 Win=65280 Len=0			
15:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 [TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0			
15:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 [TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0			
85:8b2e:e56...	2603:1061:14:01:11	TCP	86 14052 → 443 [SYN] Seq=0 Win=0 MSS=1440 WS=256 SACK_PERM			
1:14:81:11...	2405:2015:8b2e:e56...	TCP	86 443 → 14052 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=512			

Frame 387: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{1E6B850...}

Ethernet II, Src: CloudNetwork_ab8c:cb (50:c2:e8:ab:8c:cb), Dst: Sercomm_c1:76:e9 (18:fd:cb:ac:76:e9)

Internet Protocol Version 6, Src: 2405:2015:8b2e:e56d:520b:847f:5ed5, Dst: 2a04:4e42:fea1:684

Transmission Control Protocol, Src Port: 22624, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

Packets: 10976 - Displayed: 6984 (63.6%) Profile: Default

Capturing from Wi-Fi

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tcp.port==80

No.	Time	Source	Destination	Protocol	Length	Info
387.9.864892330	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0		
388.9.865063600	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 22626 → 80 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0		
392.9.873725200	2a04:4e42:fea1:684	2405:2015:8b2e:e56...	TCP	74 80 → 22626 [ACK] Seq=1 Ack=2 Win=427 Len=0		
393.9.873725600	2a04:4e42:fea1:684	2405:2015:8b2e:e56...	TCP	74 80 → 22626 [FIN, ACK] Seq=1 Ack=2 Win=427 Len=0		
394.9.873811400	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 22626 → 80 [ACK] Seq=2 Ack=2 Win=255 Len=0		
398.10.179808000	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 [TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0		
410.10.207854400	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 [TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0		
485.11.903424700	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 [TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0		
630.14.397968000	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74 [TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0		
733.17.583466900	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	86 8573 → 80 [SYN] Seq=0 Win=0 MSS=1440 WS=256 SACK_PERM		
734.17.590135500	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	86 80 → 8573 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192		
735.17.590232000	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	74 8573 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0		
736.17.590441900	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	HTTP	149 GET / HTTP/1.1		
737.17.597886000	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	74 80 → 8573 [ACK] Seq=1 Ack=76 Win=131072 Len=0		
738.17.601600000	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	943 80 → 8573 [PSH, ACK] Seq=1 Ack=76 Win=131072 Len=869 [TCP PDU reassembled in 739]		
739.17.601601200	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	HTTP	79 HTTP/1.1 200 OK (text/html)		
740.17.601675000	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	74 8573 → 80 [ACK] Seq=76 Ack=875 Win=64512 Len=0		
741.17.601869400	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	74 8573 → 80 [FIN, ACK] Seq=76 Ack=875 Win=64512 Len=0		
745.17.609809000	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	74 80 → 8573 [FIN, ACK] Seq=875 Ack=77 Win=131072 Len=0		

Frame 387: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{1E6B850...}

Ethernet II, Src: CloudNetwork_ab8c:cb (50:c2:e8:ab:8c:cb), Dst: Sercomm_c1:76:e9 (18:fd:cb:ac:76:e9)

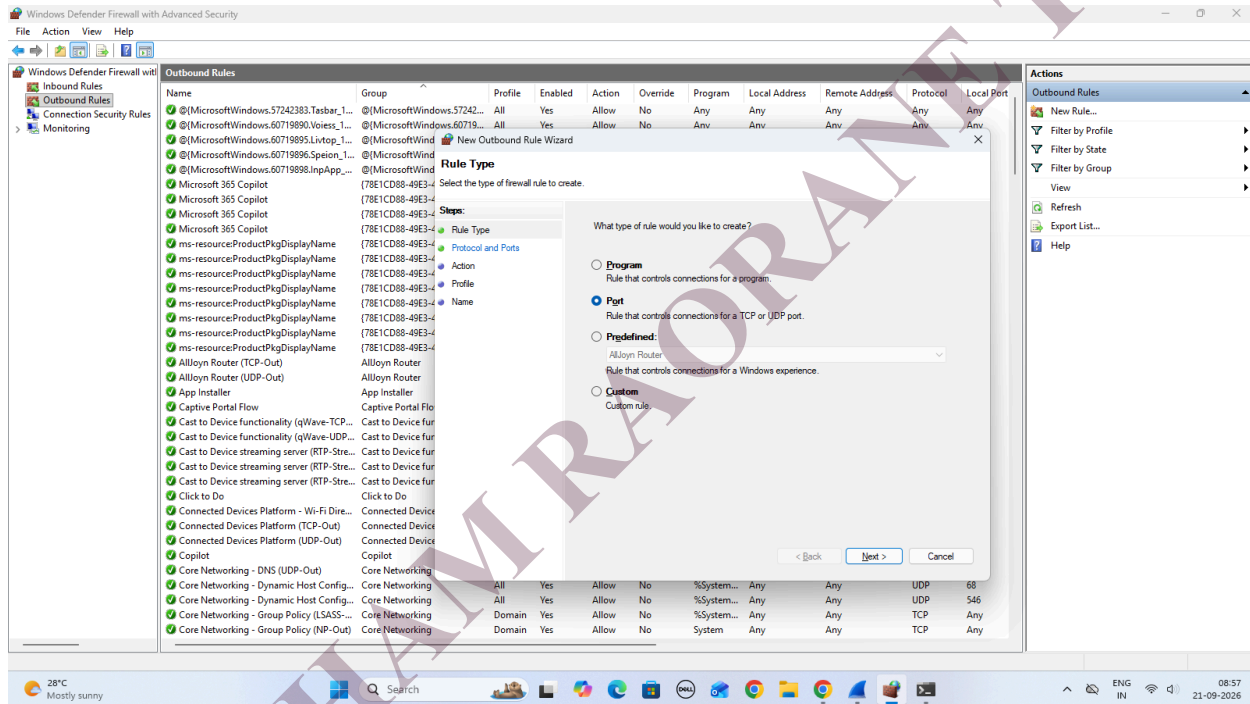
Internet Protocol Version 6, Src: 2405:2015:8b2e:e56d:520b:847f:5ed5, Dst: 2a04:4e42:fea1:684

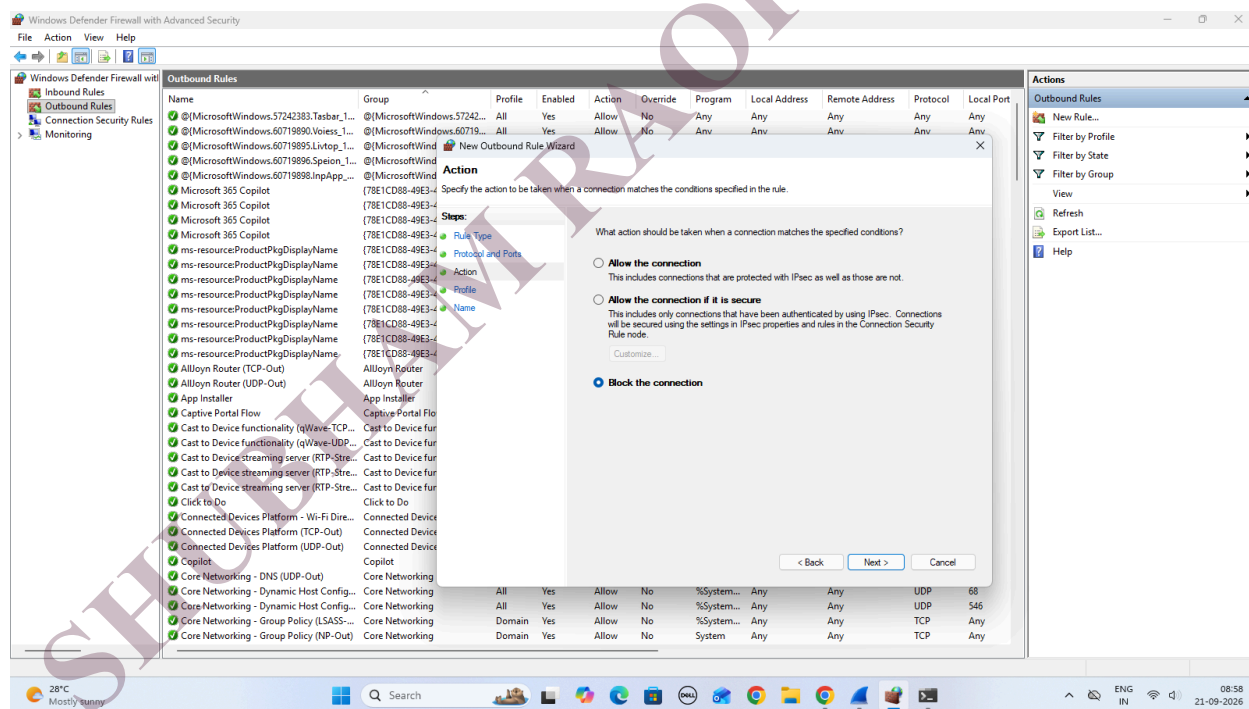
Transmission Control Protocol, Src Port: 22624, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

Packets: 11809 - Displayed: 91 (0.8%) Profile: Default

6. Create an Outbound Rule

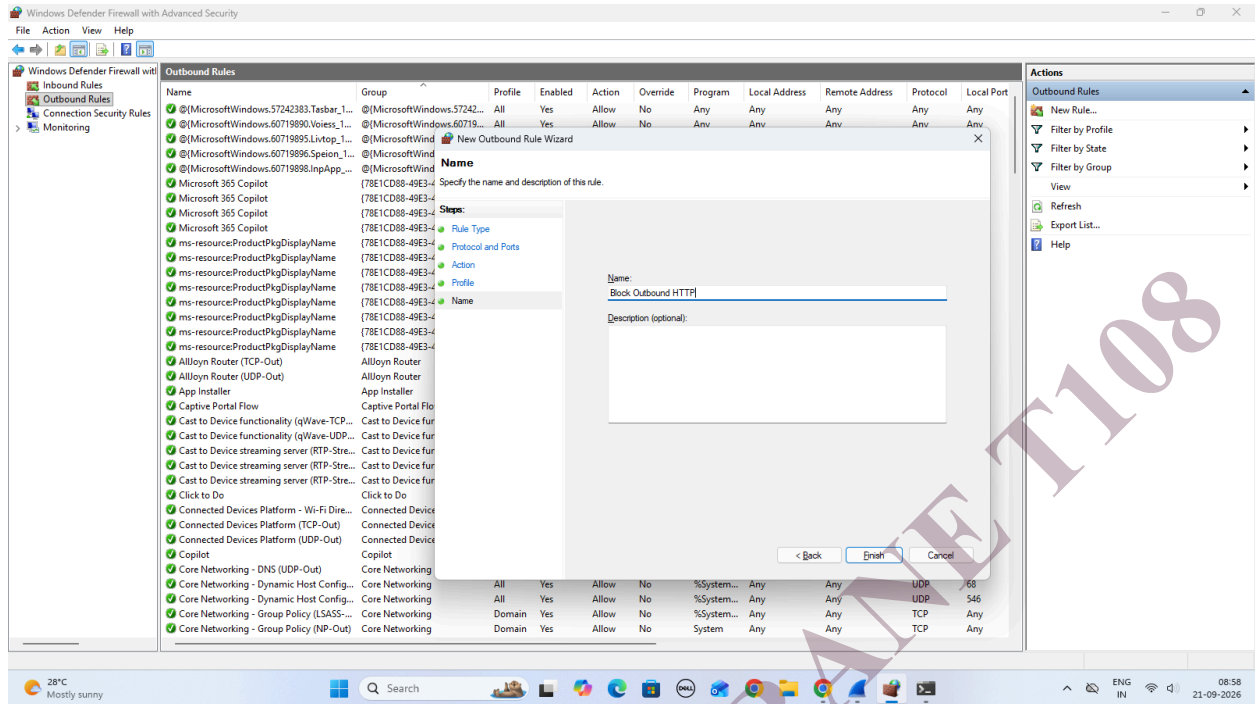
1. Open **Windows Defender Firewall with Advanced Security**.
2. Select **Outbound Rules**.
3. Click **New Rule...**
4. Select **Port**.
5. Select **TCP**.
6. Enter: **80**
7. Select **Block the connection**.
8. Select all profiles.
9. Name it: **Block Outbound HTTP**



[illegible]

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7. Observe in Wireshark

In Wireshark, use: `tcp.port == 80`

You can observe TCP packets such as:

1. SYN
2. SYN, ACK
3. ACK

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
387	9.864892300	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
388	9.865063600	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	22626 → 80 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0
392	9.873725200	2a04:4e42:fea1:684	2405:2015:8b2e:e56...	TCP	74	80 → 22626 [ACK] Seq=1 Ack=2 Win=427 Len=0
393	9.873725600	2a04:4e42:fea1:684	2405:2015:8b2e:e56...	TCP	74	80 → 22626 [FIN, ACK] Seq=1 Ack=2 Win=427 Len=0
394	9.873811400	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	22626 → 80 [ACK] Seq=2 Ack=2 Win=255 Len=0
396	10.379208800	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
410	10.787854400	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
485	11.993424700	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
630	14.397609000	2405:2015:8b2e:e56...	2a04:4e42:fea1:684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
733	17.583466900	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	86	8573 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
734	17.590135500	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	86	80 → 8573 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=0192
735	17.590233000	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	74	8573 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
736	17.590441900	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	HTTP	149	GET / HTTP/1.1
737	17.597886800	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	74	80 → 8573 [ACK] Seq=1 Ack=76 Win=131072 Len=0
738	17.601600000	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	943	80 → 8573 [PSH, ACK] Seq=1 Ack=76 Win=131072 Len=869 [TCP PDU reassembled in 739]
739	17.601601200	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	HTTP	79	HTTP/1.1 200 OK (text/html)
740	17.601675800	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	74	8573 → 80 [ACK] Seq=76 Ack=875 Win=64512 Len=0
741	17.601869400	2405:2015:8b2e:e56...	2606:4700:8d75:72db...	TCP	74	8573 → 80 [FIN, ACK] Seq=76 Ack=875 Win=64512 Len=0
745	17.609809800	2606:4700:8d75:72db...	2405:2015:8b2e:e56...	TCP	74	80 → 8573 [FIN, ACK] Seq=875 Ack=77 Win=131072 Len=0

> Frame 387: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF... (18:6B50:00000000)

> Ethernet II, Src: CloudNetwork_ab8c1cb (50:c2:e8:ab:8c:1cb), Dst: Sercomm_c76:e9 (18:fd:cb:ac:76:e9)

> Internet Protocol Version 6, Src: 2405:2015:8b2e:e56d:520b:847f:5ed5, Dst: 2a04:4e42:fea1:684

> Transmission Control Protocol, Src Port: 22624, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

wireshark-Wi-Fi\YU8V3.pcapng

Packets: 19138 - Displayed: 94 (0.5%)

Profile: Default

28°C Mostly sunny

Search

ENG IN

08:59 21-09-2026